



University of Idaho

Department of Computer Science

CS 4622/5622
Applied Data Science
with Python

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Lecture 1

A Short History and Current State of Artificial Intelligence

(not required for quizzes or assignments)

Lecture Overview

- Artificial Intelligence vs. Machine Learning vs. Deep Learning vs. Data Science
- AI approaches and goals
- Evolutional context
- Timeline of Artificial Intelligence
- Current AI capabilities
 - DL success in Computer Vision
 - DL success in Natural Language Processing
 - Large Language Models
 - Text-to-image models
- Modern AI systems
 - Foundation models
 - Reasoning models
 - AI Agents
- AI limitations and challenges
- Prospective trends in AI



Artificial Intelligence

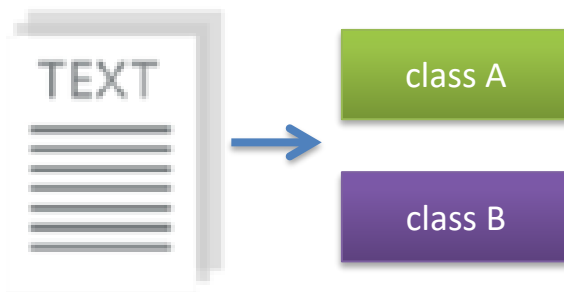
AI vs. Machine Learning vs. Deep Learning vs. Data Science

- **Artificial Intelligence (AI)** is a scientific field concerned with the development of algorithms that allow computers to reason or learn without being explicitly programmed
 - AI is the opposite to **natural (biological) intelligence** displayed by humans and animals
- AI as an academic discipline was founded in 1956
- AI studies theories and technologies related to:
 - Planning and reasoning
 - Knowledge representation
 - Machine learning
 - Natural language processing
 - Perception
 - Motion and manipulation

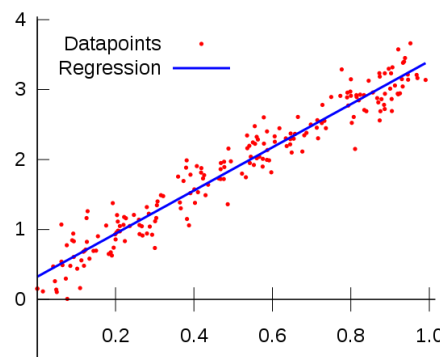
Machine Learning

AI vs. Machine Learning vs. Deep Learning vs. Data Science

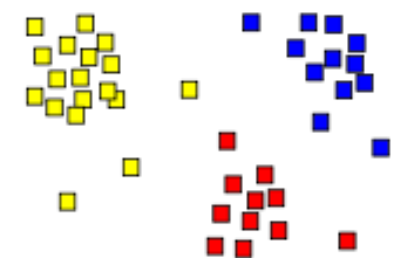
- **Machine Learning (ML)** is a subfield of Artificial Intelligence, that studies methods that learn from data and make predictions on unseen data
- Categories of ML approaches
 - **Supervised learning**: learning with **labeled data**
 - Example: image classification, email classification
 - Example: regression for predicting real-valued outputs
 - **Unsupervised learning**: discover patterns in **unlabeled data**
 - Example: cluster similar data points
 - **Reinforcement learning**: learn to act based on **feedback/reward**
 - Example: learn to play Go or Minecraft



Classification



Regression

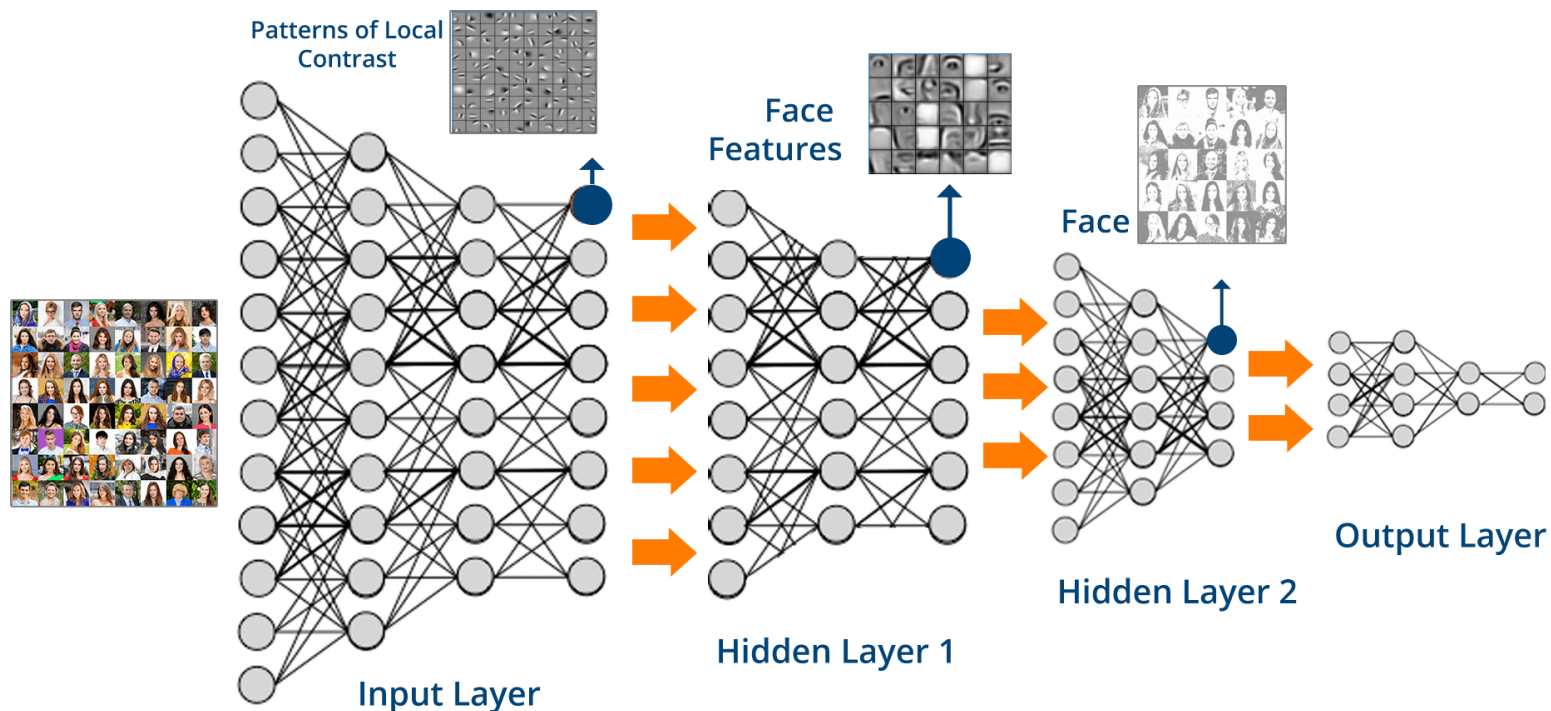


Clustering

Deep Learning

AI vs. Machine Learning vs. Deep Learning vs. Data Science

- **Deep Learning (DL)** is a sub-area in Machine Learning that uses **artificial neural networks** (ANNs) with multiple layers for learning data representations
 - Advantages of DL: ability to automatically extract features in data, processing complex high-dimensional data, scalable with data, model size, and computational power
 - The most common architectures in deep ANNs are: multi-layer perceptron NNs, convolutional NNs, recurrent NNs, graph NNs, transformer NNs

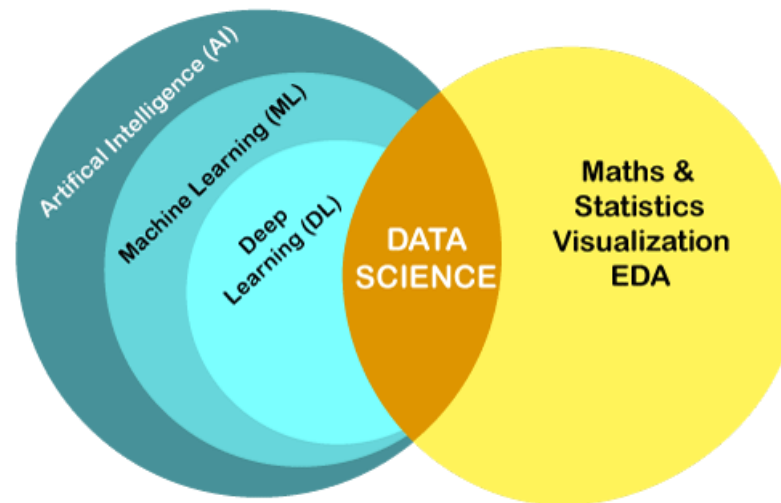


Data Science

AI vs. Machine Learning vs. Deep Learning vs. Data Science

- **Data Science (DS)** is an interdisciplinary field that uses scientific methods and algorithms to extract knowledge from data, and applies the insights to application domains (such as to make business decisions)
- Data Science versus Machine Learning
 - DS has overlaps with ML (as well as with AI and DL)
 - DS can rely on ML approaches, but it can also obtain insights from data via mathematical and statistical analysis, data visualization techniques, exploratory data analysis (EDA), or other approaches that don't necessarily require training an ML model

AI vs. ML vs. DL vs. DS



What is Intelligence?

AI Approaches and Goals

- An **intelligent agent** is any system that perceives the environment and takes actions to maximize the chances of achieving its goals
 - Goals can vary, e.g., human goals can be to make a coffee, build a wall, solve a math problem, drive a car, cook a meal, etc.
- **Definition:** *Intelligence* is an agent's ability to achieve goals in a wide range of environments
- Intelligent agents should be able to acquire and retain knowledge, and use it to respond effectively to new tasks or act in new situations and environments
 - E.g., more intelligent humans should be able to solve many physics problems that they haven't seen before (e.g., think of Einstein)
- Intelligence encompasses many related abilities for:
 - Reasoning and rational thinking, comprehending ideas, applying planning, problem-solving
 - Learning and adaptation, dealing with unexpected situations and uncertainties
 - Interacting with the real world to perceive, understand, and act

How to Develop Intelligent Machines?

AI Approaches and Goals

- AI scientists in 1950s believed that machines with human-level intelligence can be developed within 10 to 20 years
- *Initial AI approaches* included:
 - Imitate step-by-step reasoning that humans use to solve a problem
 - Create a knowledge database based on human domain knowledge about a task, and develop an inference engine to process the states and make decisions
- Challenges: handling uncertainties, combinatorial explosion (the space of solutions quickly becomes too large for most problems)
 - These approaches failed to deliver, as the scientists underestimated the complexity of human intelligence
- Various **misconceptions** about intelligence have perpetuated in the AI field
 - E.g., computers can process information -> human thinking is similar to logic processing -> encoding human thinking into a program can lead to intelligent machines
 - E.g., chess is a game of intellect and chess players are very intelligent people -> developing computers that can reason and play chess at a human expert level can lead to machines with human-level intelligence

Weak vs. Strong AI

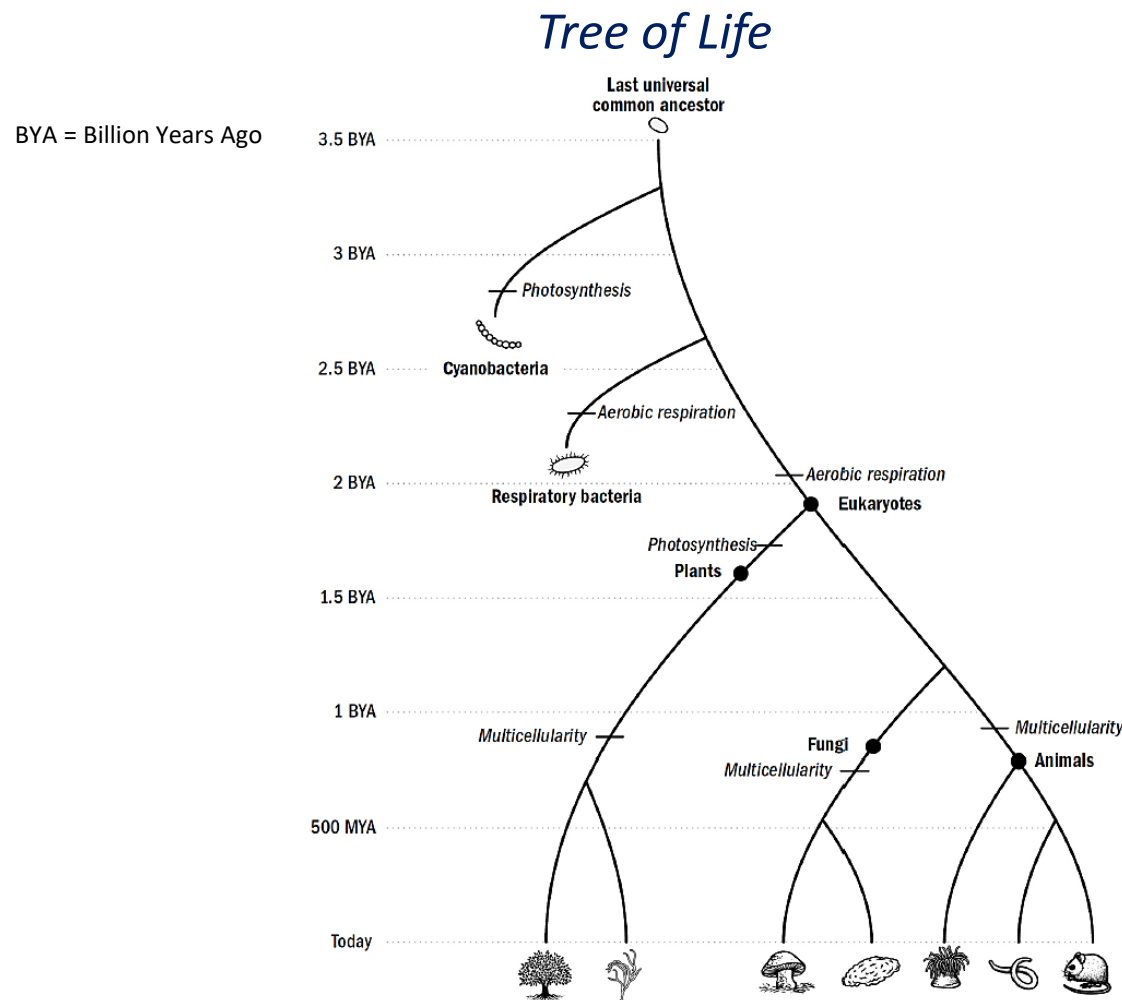
AI Approaches and Goals

- AI systems can be classified into weak AI and strong AI systems
- *Weak AI*, or *narrow AI*: can solve one specific task
 - E.g., image classification ML models
 - E.g., Deep Blue computer that defeated the world chess champion
- *Strong AI*, or *artificial general intelligence (AGI)*: can solve a variety of tasks
 - AGI is the ability to understand or learn any intellectual task that a human being can
 - AGI performance would be indistinguishable from that of humans
 - At present, AGI systems do not exist
 - Predictions for achieving AGI vary widely among experts, ranging from within 5 years to more than 50 years

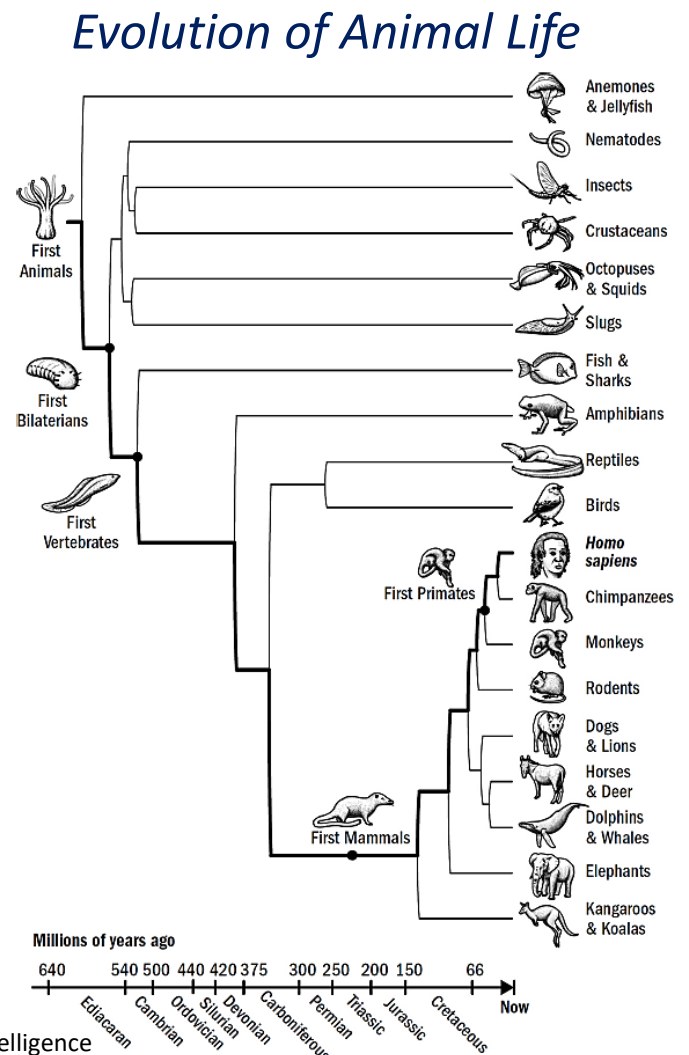
Timeline of Biological Intelligence

Evolutional Context

- Animals exhibit intelligent behavior: they learn and adapt, and plan their actions to achieve goals



Source: Max Bennett (2023) A Brief History of Intelligence



Timeline of Biological Intelligence

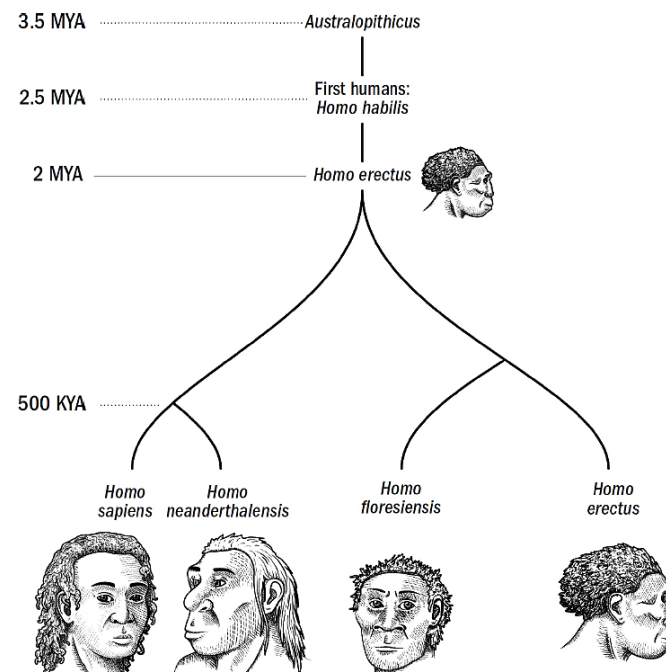
Evolutional Context

- Early humans inherited brain structures and features of intelligence from our evolutionary ancestors
 - The human brain is a scaled-up primate brain, without new neurological systems
 - The difference from the brain of a mouse involves only a few brain modifications
 - The brain of a fish has almost all basic structures as our brain
 - Our larger-sized brain enabled the development of language, abstract reasoning, long-term planning, and other advanced cognitive abilities, superior to that of primates and other animals

The First Humans

MYA = Million Years Ago

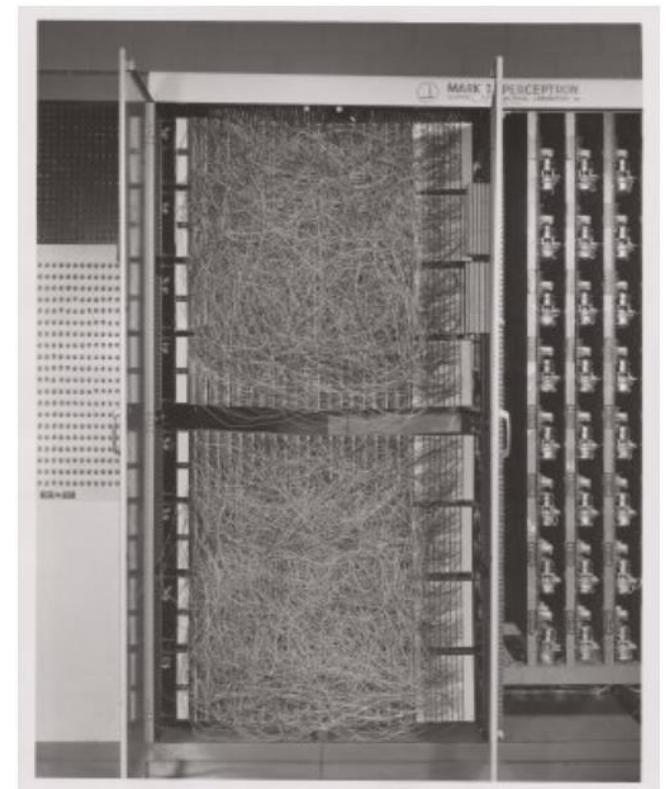
KYA = Thousand Years Ago



AI Timeline: Early AI

AI Timeline

- 1943 – The first model of a simple artificial neuron proposed
- 1950 – Alan Turing introduced the **Turing test**
- 1955 – The **term Artificial Intelligence** used for the first time
- **1956 – Workshop on AI held in Dartmouth College**, New Hampshire, organized by John McCarthy, Marvin Minsky, Nathaniel Rochester, Claude Shannon
 - Official beginning of AI as academic discipline
 - A statement from the workshop proposal: "Every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it."
- 1958 – **Perceptron** algorithm proposed by Rosenblatt
 - Shown is the Mark I Perceptron computer, used for implementing the algorithm



AI Timeline: Early AI

AI Timeline

- 1958 – The term **Machine Learning** was used for the first time
- 1961 – The first industrial **robot** Unimate was installed on an assembly line at General Motors in New Jersey
- 1966 – **Eliza**, a chatbot that simulates conversations with a psychotherapist
- 1970-1980 – First **AI winter**, agencies reduced funding for AI projects due to unsatisfactory progress
- 1982 – An expert systems deployed for configuring computer orders
- 1987-1992 – Second **AI winter**, DARPA cut AI funding for expert systems
- 1995 – The advent of machine learning and statistical methods
- 1997 – IBM's supercomputer **Deep Blue** won against world chess champion Gary Kasparov



AI Timeline: Modern AI

AI Timeline

- 2002 – Vacuum robots Roomba released
- 2006 – Facebook, Netflix, Twitter started using AI-based recommender systems
- 2011 – IBM's supercomputer **Watson** won against two human rivals in the quiz show Jeopardy
- 2012 – **Deep NN model AlexNet** won image classification contest - *beginning of the era of deep learning*
- 2015 – **GAN** (Generative Adversarial Network) introduced
- 2016 – Google's DeepMind program **AlphaGo** defeated the Go grandmaster Lee Sedol
 - The game of Go is more difficult than chess, because the number of possible moves is much greater



AI Timeline: Recent AI

AI Timeline

- 2017 – **Transformer** network architecture was introduced in the paper by Vaswani et al. “Attention Is All You Need”
- 2020 – OpenAI’s **GPT-3** is the first large language model with 175B parameters, performed well on many NLP tasks
- 2021 – DeepMind’s **AlphaFold** achieved high accuracy in predicting the 3-dimensional shape of proteins
- 2022 – OpenAI’s **DALL·E 2** generated photorealistic images with remarkable quality
- 2022 – Facebook’s **NLLB** (No Language Left Behind) model for machine translation between 200 languages
- 2022 – OpenAI released **ChatGPT**, a large language model with human-like abilities in answering questions and chatting
- 2023 – Meta AI’ **Llama 2** is the first open-source large language model that is freely available for commercial use

AI Timeline: Recent AI

AI Timeline

- 2023 – Mistral released Mixtral LLM, a **mixture-of-experts model** that integrates 8 LLMs, and outperformed larger models on standard benchmarks
- 2023 – Google launched **Gemini**, Anthropic launched **Claude 3**: both reached the top spot on standards benchmarks at the time of launching
- 2024 – **Vision-Language models** (GPT-4 omni, LlaVA, Claude) for image interpretation and analysis
- 2024 – **Multi-Agent frameworks** (LangChain, AutoGPT, AutoGen) connect to other services and provide assistance in making reservations, planning a trip, etc.
- 2025 – **Turing test** passed by GPT-4.5, it was mistaken for human 73% of the time
- 2025 – GPT-o3 and Gemini Deep Think won **gold medal** at the International Mathematical Olympiad (IMO)

DL Success in Computer Vision

Current AI Capabilities

- *Computer Vision* tasks
 - Image and video recognition/classification, segmentation, object detection, image synthesis
- Important architectures in CV
 - AlexNet – 2012
 - Convolutional NNs for image recognition, 5 layers, GPU for parallel processing
 - ImageNet Large Scale Visual Recognition Challenge (ILSVRC): AlexNet reduced the error on ImageNet from 26% by traditional ML approaches to 15%
 - VGG – 2014
 - 16 layers CNN architecture
 - Inception – 2015
 - Stacked 1x1 convolutions, 22 convolutional layers
 - ResNet – 2015
 - Introduced residual connections, it is a family of networks with 18, 34, 50, 101, and 152 layers
 - Several related models were proposed afterwards, e.g., ResNeXt (2017), EfficientNet (2019)
 - Vision Transformers (ViT) – 2020
 - Employ attention layers, inspired by the transformer models used in NLP

DL Success in Natural Language Processing

Current AI Capabilities

- *Natural Language Processing* (NLP) tasks
 - Text classification, text summarization, speech recognition, machine translation, dialog generation, part-of-speech tagging
- In the last decade, *Large Language Models (LLMs)* powered by transformer NNs achieved unprecedented success in NLP tasks
 - **Training data** for LLMs is a diverse set of text data gathered from the web
 - E.g., text collected from Wikipedia, e-books, news articles, and many other sources
 - Data preprocessing: words are projected into an embeddings space, where each word is replaced with a numerical **token**
 - Given a sequence of words (tokens) from a dictionary, LLMs are trained to predict the next word (i.e., assign the probability of the next token)

Input: A quick brown	Output: fox
Input: Marry had a little	Output: lamb
Input: Nothing is	Output: impossible
 - To adapt LLMs for specific tasks, the trained models are typically **fine-tuned** on smaller, specific datasets



Large Language Models

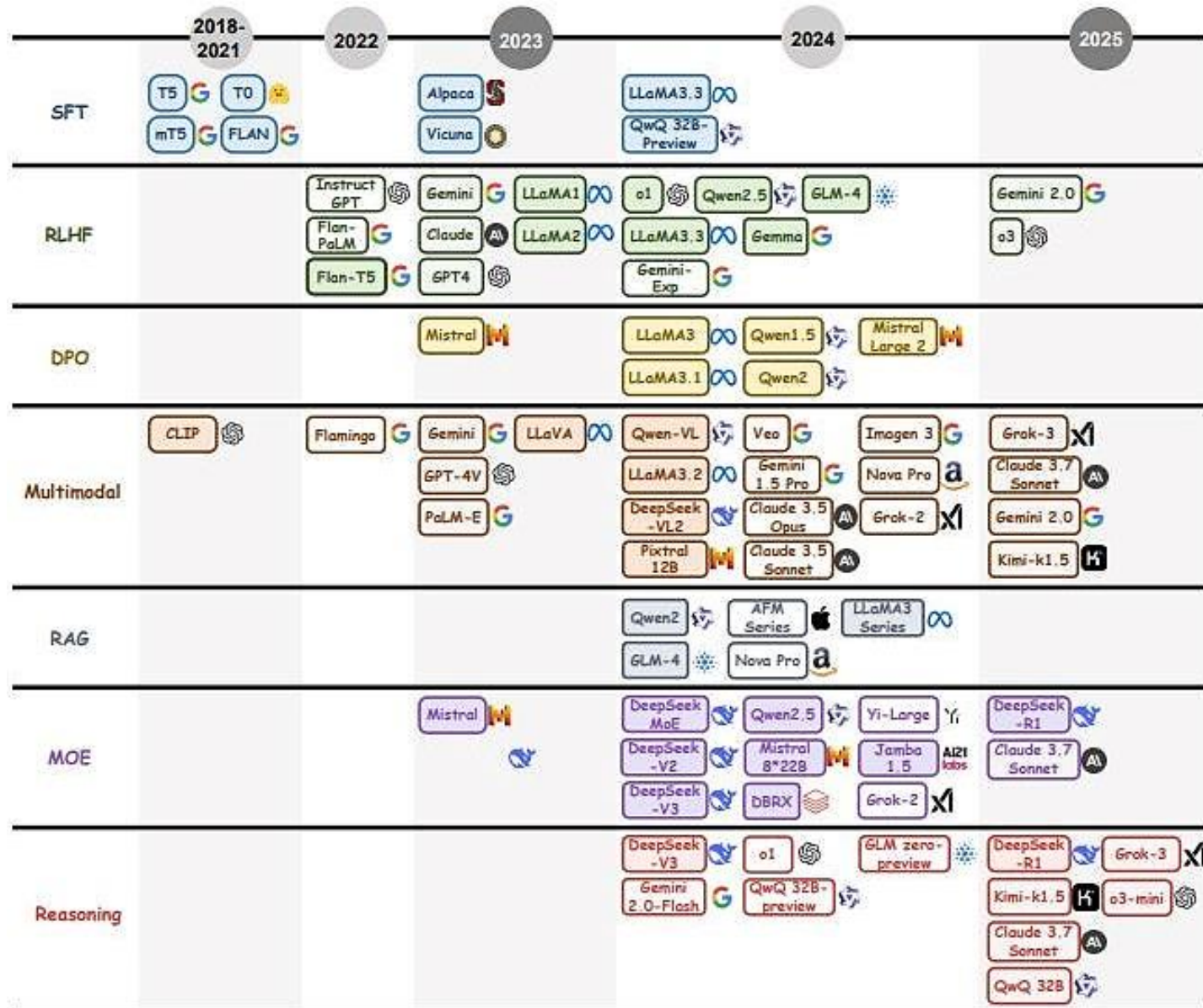
Current AI Capabilities

- Recent LLMs are very large deep neural networks having trillions of parameters
 - GPT-4: OpenAI, 1.76T (trillion) parameters
 - Gemini 2.5 Pro: Google, 1.5T parameters
 - Claude 4 Opus: Anthropic, 2T parameters
 - Llama 3.1: Meta AI, 405B parameters
 - Note that compared to the human brain having between 100 and 500 trillion synaptic connections, current LLM models are still fairly small
- Training LLMs requires substantial *computational resources* and time
 - Typically tens of thousands of GPUs are used for LLM trained
 - E.g., purchasing cost of A100 GPU is about \$15K -> cost of 10,000 A100 GPUs is about \$150M
 - E.g., purchasing cost of H200 GPU is about \$50K -> cost of 10,000 H200 GPUs is about \$500M
- *Concerns* regarding LLMs
 - Misuse and unethical use of AI, amplifying disinformation, environmental impact (high carbon emissions), increasing economic inequalities, centralization of power (e.g., affordable only by the largest corporations)



Timeline of Large Language Models

Current AI Capabilities





LLMs Leader Board by LMSYS

Current AI Capabilities

Rank (UB) ↑	Model ↓	Score ↓	95% CI (±) ↓	Votes ↓	Organization ↓	License ↓
1	 gpt-5-high	1481	±11	3,181	OpenAI	Proprietary
2	 gemini-2.5-pro	1458	±5	28,091	Google	Proprietary
2	 o3-2025-04-16	1451	±5	34,027	OpenAI	Proprietary
2	 claude-opus-4-1-20250805	1446	±9	5,187	Anthropic	Proprietary
4	 chatgpt-4o-latest-20250326	1440	±5	32,125	OpenAI	Proprietary
4	 gpt-4.5-preview-2025-02-27	1438	±6	15,271	OpenAI	Proprietary
5	 grok-4-0709	1430	±6	14,078	xAI	Proprietary
5	 qwen3-235b-a22b-instruct-2507	1428	±8	6,097	Alibaba	Apache 2.0
7	 kimi-k2-0711-preview	1420	±6	13,578	Moonshot	Modified MIT
7	 claude-opus-4-20250514-thinking-16k	1420	±6	19,330	Anthropic	Proprietary
8	 deepseek-r1-0528	1418	±6	19,389	DeepSeek	MIT
8	 glm-4.5	1414	±8	5,907	Z.ai	MIT
9	 claude-opus-4-20250514	1412	±5	27,428	Anthropic	Proprietary

Text-to-Image/Text-to-Video Models

Current AI Capabilities

- Text-to-image and text-to-video models produce images/videos with remarkable photorealism, accurate fine details, compositionally, spatial relations of the objects, and even with creativity in the synthesis
 - These models typically employ **diffusion probabilistic networks**, which learn the steps of adding and removing noise to images
- *Text-to-image models*
 - DALL·E 3 (OpenAI)
 - Imagen (Google)
 - Stable Diffusion (Stability.ai)
 - Midjourney (Midjourney.com)
- *Text-to-video models*
 - Sora (OpenAI)
 - Veo (Google)
 - Dream Machine (Luma Labs)
 - Runway (RunwayAI)

Images Generated by DALL·E 2

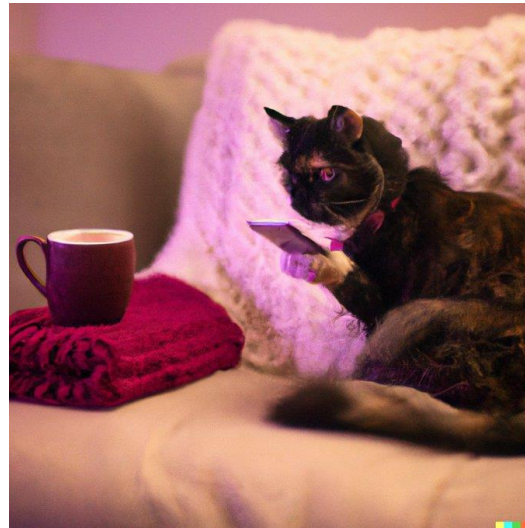
Current AI Capabilities

- These are a few (cherry-picked) examples of images generated by DALL·E 2

A photo of a quaint flower shop storefront with a pastel green and clean white facade and open door and big window



Cat sipping tea and posting to twitter while sitting on a couch



A rabbit detective sitting on a park bench and reading a newspaper in a victorian setting



A lion in a hoodie hacking on a laptop



Teddy bears shopping for groceries in ancient Egypt



Teddy bears working on new AI research on the moon in the 1980s



Foundation Models

Modern AI Systems

- **Foundation models** are large NN models trained at **scale** with high capabilities for **transfer learning** to many other applications
- The scale of these models results in new **emergent capabilities** – e.g., they perform well on tasks on which they were not explicitly trained to do
 - “**Emergence** is when quantitative changes in a system result in qualitative changes in behavior”
 - This allow fine-tuning to new tasks with a small number of training data instances
 - E.g., **few-shot learning** refers to fine-tuning with only a few instances
- Notable applications of pretrained LLMs include:
 - Programming code completion models: CoPilot, AlphaCode, Codex, Codegen
 - Text-to-image generative models: DALL·E, Imagen, Stable Diffusion
 - Protein sequence prediction, solving math problems, preparing legal documents (other task examples are listed on the next page)
- Transfer learning is what makes foundation models possible, but scale is what makes them powerful

Foundation Models

Modern AI Systems

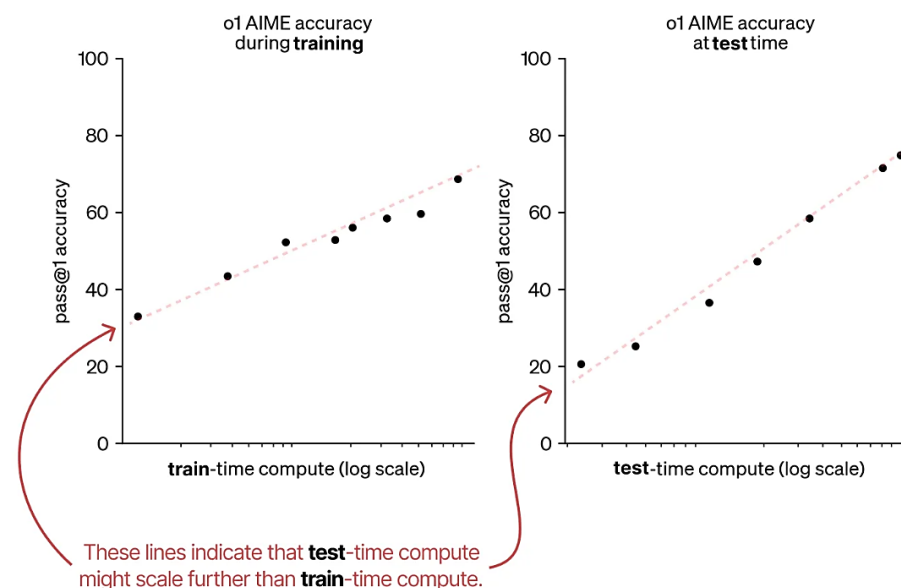
- Examples of applications and downstream tasks in which foundations models are being used

Program writing	Image captioning	Generate images	Parse data	Classify text
Use natural language to generate SQL/Python/Java code	Describe and classify images	Create images based on natural language	Extract data from images	Identify entities, parts-of-speech, and other text categories
Q&A	Writing assistant	Summarize	Solve homework	Translate
Answer natural language questions based on knowledge base	Correct your writing	Summarize text to key concepts	Solve basic math and programming problems	Translate text from one language to another
Code explanation	Copywriting	Sentiment rating	Recipe creation	Chat
Writes the description of code functionality in natural language	Generate ad/product/job descriptions based on short prompts	Rates the sentiment, toxicity, warmth, etc. of text	Use at your own risk	Talks like a human

LLM Scaling Laws

Modern AI Systems

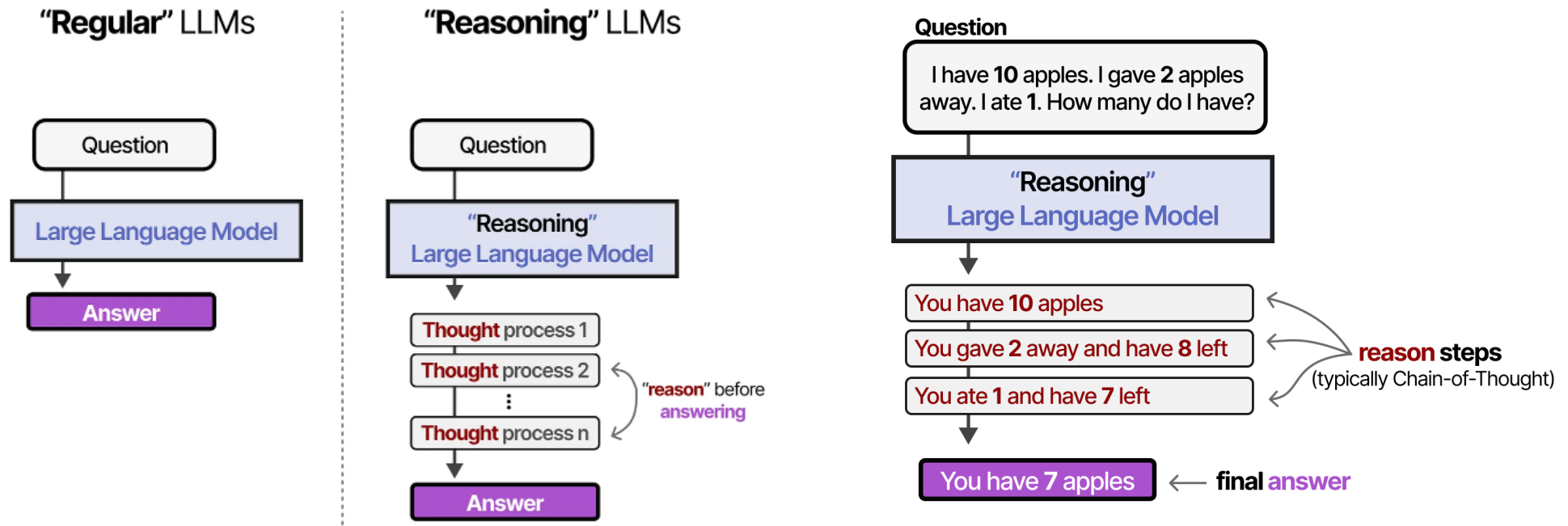
- LLM *scaling laws* provide a relationship between the model's performance and model's scale (typically shown on a log scale)
 - Well known **train-time compute scaling laws** include Chinchilla and Kaplan laws (see left figure below)
 - They indicate that performance increases with more compute, tokens, and parameters
 - OpenAI's **test-time compute scaling law** (right figure below) suggests that test-time compute follows the same trend as train-time compute
 - Performance improves by allowing LLMs to “think longer” during inference



Reasoning LLMs

Modern AI Systems

- **Reasoning LLMs** perform multiple reasoning steps before providing an answer at test time
 - The **reasoning steps** or **thought process** break down the inference process into smaller, structured inferences (**Chain-of-Thought**)
 - Reasoning LLMs include: OpenAI o1 and o3, DeepSeek-R1, Google Gemini 2.0 Flash Thinking, and actually most of the latest LLMs have reasoning abilities





AI Agents

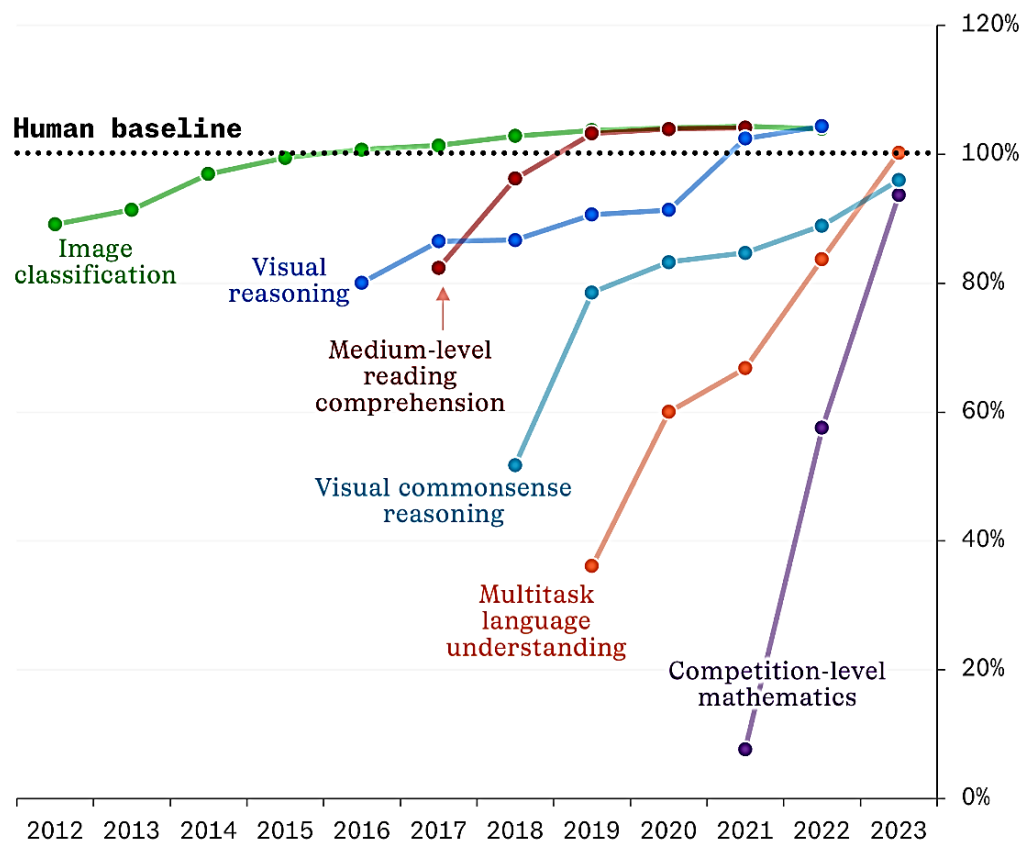
Modern AI Systems

- *AI Agents* use LLMs or other resources to understand instructions, reason through tasks, and take actions across digital tools or environments
- Abilities
 - Interpret natural language commands
 - Plan multi-step solutions
 - Call external tools (e.g., web search, code execution)
 - Adapt behavior based on feedback
- Examples of use
 - Automating document analysis
 - Running complex simulations
 - Managing workflows (e.g., email sorting, report writing)
 - Scientific discovery and literature review

Progress in AI

Modern AI Systems

- The graph shows a comparison between AI performance and human baseline performance (100%) on benchmarks dataset for several tasks
 - E.g., handwritten recognition is evaluated on MNIST dataset, image recognition is based on ImageNet dataset, etc.



Evolution of AI Capabilities

Modern AI Systems

- **Pattern recognition period**
 - Analyze existing data, make predictions
 - Classification models for identifying spam emails, recognizing handwritten digits
 - Regression models for predicting house prices, stock trends
- **Generative AI period**
 - Generate new content
 - LLMs for generating human-like text and conversations
 - Text-to-image models for creating realistic images from text prompts
 - Models for composing music, writing code, creating videos
- **Task execution AI period**
 - AI agents perform actions
 - Booking flights and making reservations
 - Analyzing documents and writing reports
 - Controlling software applications
- **Next AI period (?)**
 - Multiple AI agents working together, integration with physical robots



Engineering vs Science Phase of Technology

AI Limitations and Challenges

- Theoretical guarantees about AI performance are lacking at the present time
 - Currently, AI is in *Engineering phase*: models are designed to solve tasks, are integrated into new products, add value to companies, have economic impact
 - *Science phase* of AI is to follow: theory is developed to guarantee convergence, prove stability, or explain successes and failures of models
- Various technologies historically began with an engineering phase (inventions made, products built) to be later followed by a science phase (theory developed)
 - Steam engines were used in paper mills and factories since 1776; the theory of Thermodynamics was developed between 1820s and 1850s
 - Airplanes were constructed and flown since 1904; the modern theory of Aerodynamics was developed in 1930s
 - Electric circuits were discovered around 1800; the theory of Electromagnetism was founded between 1820s and 1830s

Trustworthy AI

AI Limitations and Challenges

- **Trustworthy AI** – efforts to address the limitations to ensure that end-users can trust the predictions by AI models
- Topics in trustworthy AI include:
 - **Robustness**
 - Even unnoticeably small perturbations can impact the model predictions
 - **Generalization**
 - OOD (out-of-distribution) inputs; e.g., a model trained on medical images in one hospital performs poorly on images in another hospital (due to different equipment or settings used)
 - **Explainability**
 - The decision-making process of large models is non-transparent and difficult to understand
 - **Fairness**
 - Predictions can show bias against demographic groups, based on gender, age, culture
 - **Privacy protection**
 - Models can memorize and reveal input data; e.g., a model can reveal sensitive private information in medical records used for training
 - **Ethics**
 - The models should produce ethical decisions that are aligned with our human values (also referred to as **AI Alignment**)



Moravec's Paradox

AI Limitations and Challenges

- The paradox is based on observations by the researcher **Hans Moravec**: the hard problems for humans are easy for computers, and the easy problems for humans are hard for computers
 - E.g., it is easier for computers to play chess or solve logic problems, than to make a coffee or plant a tree
 - Robotic tasks that involve sensorimotor and perception skills are more difficult than planning and reasoning tasks
- One possible explanation of Moravec's paradox is because sensorimotor skills are older and developed earlier in the evolution than reasoning skills
 - We can perform skills like recognizing a face, moving in space, or catching a ball unconsciously and effortlessly, because they have been optimized over millions of years
 - Abstract thinking, reasoning, or solving math problems are thousands of years old skills that we haven't mastered completely, and we need to put more effort
- We should expect that the difficulty to automate a skill with AI is proportional to the amount of time it took for the skill to develop in animals and humans

The Bitter Lesson

AI Limitations and Challenges

- *The Bitter Lesson* (2019) is a short paper by Rich Sutton
 - <http://www.incompleteideas.net/IncIdeas/BitterLesson.html>
- The Bitter Lesson is based on his observations regarding the historical development of AI methods, which can be characterized with three phases:
 - Phase 1 - AI researchers incorporate human domain knowledge into their AI methods, which helps in short term
 - Phase 2 - In the long term, the performance of such models plateaus without further progress
 - Phase 3 - Progress is eventually achieved by general methods that scale computation with search and learning
- In conclusion:
 - AI methods that **leverage computation and search at scale** are the most effective
 - Human-centric approaches complicate methods and make them less suited to leveraging computation and search at scale
 - The search for solutions should be done by our methods, not by us
 - We need AI methods that can discover like us, and not based on our discoveries

Prospective Trends in AI

Prospective Trends in AI

- *Unsupervised/self-supervised learning*
 - Increased reliance of methods that learn from raw data without annotations or labels
- *Training/testing at scale*
 - Further scaling along the three main factors: amount of computation, number of model parameters, and training dataset size
 - Test-time compute scaling offers additional potential for performance improvement
- *Multi-modal learning*
 - Capacity to learn from multiple sources of information: text, audio, video, sensor data
 - Task-specific models replaced with general models capable of solving multiple tasks
- *Agentic frameworks*
 - Development of domain-specialized AI agents that use external tools, apply multi-step reasoning, and perform actions
- *New algorithms (e.g., reinforcement learning, neuro-symbolic learning)*
 - Advances in reinforcement learning for autonomous exploration and adaptation, methods that combine neural and symbolic reasoning, etc.